

trials, each having probability 0.5, is as follows:

```
> pbinom(5, 10, 0.5)
[1] 0.6230469
```

In addition to the common arguments listed previously, the binomial distribution function accepts the following arguments:

Arguments	Description
size	Number of trials
prob	Probability of success in each trial

Exponential

The exponential function for a given rate 'lambda' is given by:

$$f(x) = \lambda e^{-\lambda x}$$

...where $x \geq 0$. The *rexp()* function provides random deviations, the *qexp()* function gives the quantile function, and the *pexp()* function provides the distribution function. A couple of examples are given below:

```
> rexp(5, rate = 1)
[1] 0.44090566 0.18508747 1.18596321 0.86235515 0.05020871

> pexp(d, rate = 1)
[1] 0.38571907 0.23414656 0.13950812 0.54148378 0.34837687
0.05119434
[7] 0.41766053 0.42140839 0.34755644 0.27114996
```

The specific 'rate' argument is applicable to the exponential function.

Arguments	Description
rate	A vector of rates

Geometric

The geometric distribution density with probability 'p' is defined as follows:

$$p(x) = p(1-p)^x$$

...where $x = 0, 1, 2$, etc, and 'p' is between 0 and 1. The *rgeom()* function for 10 observations and probability 0.5 is given below:

```
> d <- rgeom(10, 0.5)
> d
[1] 0 1 1 0 0 0 0 0 1 1
```

The *pgeom()* and *qgeom()* functions are based on closed-form formulae and provide the distribution and quantile functions, respectively.

```
> pgeom(d, 0.5)
[1] 0.50 0.75 0.75 0.50 0.50 0.50 0.50 0.50 0.75 0.75

> qgeom((1:10)/15, prob = 0.2)
[1] 0 0 0 1 1 1 2 2 3 4
```

The argument specific to the geometric function is given below:

Arguments	Description
prob	Probability of success in each trial

Normal

The normal distribution for mean 'mu' and standard deviation 'sigma' is given by:

$$f(x) = 1/(\sqrt{2\pi}\sigma) e^{-((x - \mu)^2/(2\sigma^2))}$$

A sample of ten random values for mean 0 with standard deviation 1 can be generated using the *rnorm()* function, as follows:

```
> d <- rnorm(10, mean=0, sd=1)
> d
[1] 0.66512403 -0.43102217 0.50305386 1.55028087
1.20598557 0.33704698
[7] 0.09200192 0.67090448 -0.24424568 -0.78798191
```

Example uses of *pnorm()* and *dnorm()* functions are given below:

```
> pnorm(d, mean=0, sd=1)to
[1] 0.7470144 0.3332261 0.6925368 0.9394629 0.8860885
0.6319593 0.5366517
[8] 0.7488593 0.4035203 0.2153536

> dnorm(d, mean=0, sd=1)
[1] 0.3197763 0.3635536 0.3515265 0.1199568 0.1927928
0.3769138 0.3972575
[8] 0.3185439 0.3872184 0.2924691
```

The normal distribution specific arguments are listed for your reference below.

Arguments	Description
mean	A vector of means
sd	A vector of standard deviations